Group 5: Project Plan Zachakos, de Sousa, Willoughby, House and McArthur

Project Description:

University timetabling is a yearly battle with many hours and resources dedicated to getting it right. The University of Edinburgh Timetabling department has tasked us with exploring some avenues for future improvements to the timetable and its structure. Our group has decided tackling questions within the TIME set of scenarios that were laid out to us. Specifically, we will be focusing on a reduction in core hours, maintaining a lunch period in the time frame of 12pm-2pm, minimizing clashes and an overall reduction in the idle time for rooms across all campuses.

Targets:

1) A Reduction in core hours from 9am-6pm to 9am-5pm.
   Stretch Goal: Reducing core hours as far as possible with minimal negative impact on other areas of timetable structure and integrity.
2) Maintaining the existence of a lunch break for all students with at least 4 hours of contact time in a day.
   Stretch Goal: Lunch breaks structured in a way to minimise the overloading of facility capacities.
3) Minimizing clashes at a programme level where possible (including penalising solutions for the introduction of clashes).
   Stretch Goal: Minimizing clashes at an individual student level.
4) A reduction in idle times for rooms across all campuses, notably lectures theatres at the Kings Buildings campus are idle for much of the week.
   Stretch Goal: Have all rooms hit the 75% usage target where possible.

The success of the project will measured by the extent at which we are able to succeed in the targets laid out while the stretch goals will help to measure how far we have succeeded beyond this.

To facilitate the development of a successful model some key steps that we will be taking include:

- Initially optimizing the original timetable at its 9am-6pm cores hours initially to ease the computational burden once we begin to reduce the core hours.
- Modelling/timetabling the Veterinary and Medical Schools separately as their programmes are almost completely separated from the rest of the population.
    - This will also provide a secondary benefit of being a quick way to test the viability of our approach.

The modelling approach is likely include a combination of Constraint Programming to capture the more abstract constrains and Mixed Integer Programming to complete a bulk of the more straightforward computations with mathheuristics used in this stage to lighten this computational burden. This approach being inspired by the methodology of Lamas-Fernandez, Matrinez-Sykora and Potts (2021) [1] as part of their win in the 2021 International Timetabling Competition.

From further literature review we are also considering the implementation of a number of novel techniques. From the model presented by Aygül et al. (2025) [2] we have interest in utilising the common triples and course splitting considerations. We will also be looking into Particle Swarm Optimization as laid out by Chen and Shih (2013) [3]. More broadly the overarching class of Genetic Algorithms, Alghamdi et al (2020) [4], shows promise for efficient solving in this problem area. This is however subject to change as the project evolves and develops over its course.

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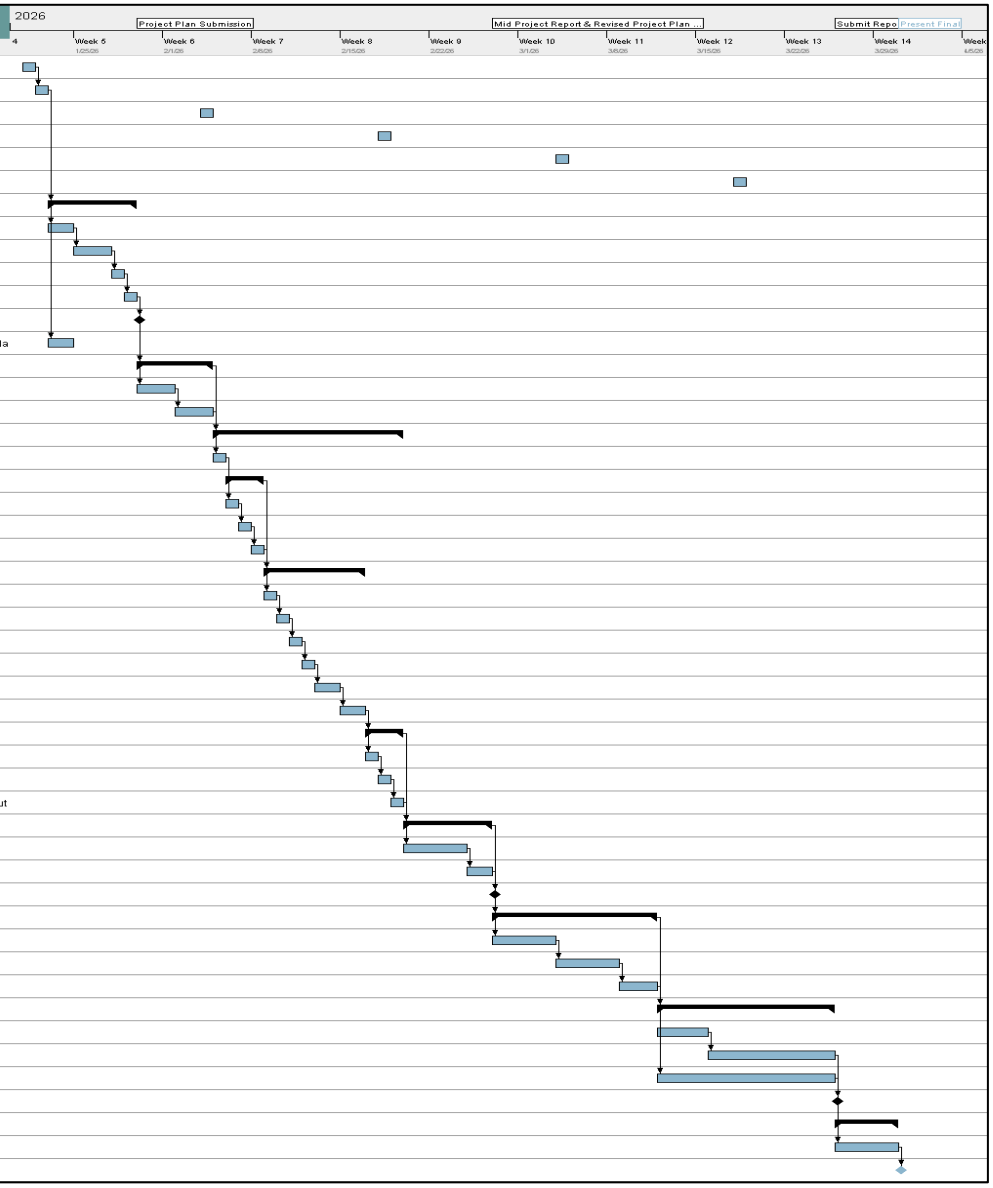
Name	Begin date	End date	Resources	Deliverables
Decide on Project Question	1/21/26	1/21/26	All	
Workshop 1	1/22/26	1/22/26	All	
Workshop 2	2/4/26	2/4/26	All	
Workshop 3	2/18/26	2/18/26	All	
Workshop 4	3/4/26	3/4/26	All	
Workshop 5	3/18/26	3/18/26	All	
Project Plan Development	1/23/26	1/29/26		
Write Project Overview	1/23/26	1/24/26	Todd	Project description
Develop Task List	1/25/26	1/27/26	Todd	Task list
Build Gantt Chart	1/28/26	1/28/26	Todd	Gantt Chart
Develop Risk Register	1/29/26	1/29/26	Todd	Risk Register
Project Plan Submission	1/30/26	1/30/26	All	Project plan report
Final Report Introduction	1/23/26	1/24/26	Nolan	Written project background and pla
Data Analysis and Subset Selection	1/30/26	2/4/26		
Assess Data for Errors	1/30/26	2/1/26	Todd,Alex,Conor	
Decide on Appropriate Subset	2/2/26	2/4/26	Alex,Conor	Assess Data for Errors
Initial Model Development	2/5/26	2/19/26		
Data Verification for Initital Model	2/5/26	2/5/26	Todd	Approved data for initial code
Write Mathematical Model	2/6/26	2/8/26		
Develop OBJ Function	2/6/26	2/6/26	Alex,Nolan,Conor	
Develop Constraints	2/7/26	2/7/26	Alex,Nolan,Conor	
Identify All Variables	2/8/26	2/8/26	Alex,Nolan,Conor	Written mathematical model
Initial Code Development	2/9/26	2/16/26		
Confirm Software Choice	2/9/26	2/9/26	Ben,Nolan	
Initialize Data	2/10/26	2/10/26	Ben,Nolan	
Write OBJ Function	2/11/26	2/11/26	Ben,Nolan	
Format Constraints	2/12/26	2/12/26	Ben,Nolan	Core code base
Trial Runs	2/13/26	2/14/26	Ben,Nolan	
Output Analysis	2/15/26	2/16/26	Ben,Nolan	Initial optimisation output
Application of Code to Broader Data	2/17/26	2/19/26		
Trial Runs	2/17/26	2/17/26	Ben,Nolan	
Iterative Application	2/18/26	2/18/26	Ben,Nolan	
Analysis and Improvements	2/19/26	2/19/26	Todd,Alex,Conor	Larger dataset optimisation output
Mid Project Report Development	2/20/26	2/26/26		
Write Report on Current Progress	2/20/26	2/24/26	Todd,Conor	
Update Project Plan	2/25/26	2/26/26	Todd	
Mid Project Report & Revised Project Plan Submission	2/27/26	2/27/26	All	Project status repor...
Goal Completion	2/27/26	3/11/26		
Focused Application of Results to Questions	2/27/26	3/3/26	Nolan	
Adustments to Refine Questions/Answers	3/4/26	3/8/26	Ben,Alex,Conor	
Begin Recommendation Alignment	3/9/26	3/11/26	Todd	Written draft recommendations
Final Project Report Development	3/12/26	3/25/26		
Compile Final Results and Recommendations	3/12/26	3/15/26	All	
Write Report	3/16/26	3/25/26	All	Final report
Build Presentation	3/12/26	3/25/26	Todd	
Submit Report	3/26/26	3/26/26	All	Written final report
Final Presentation	3/26/26	3/30/26		
Presentation Rehearsals	3/26/26	3/30/26	All	
Present Final	3/31/26	3/31/26	All	Final presentation deck



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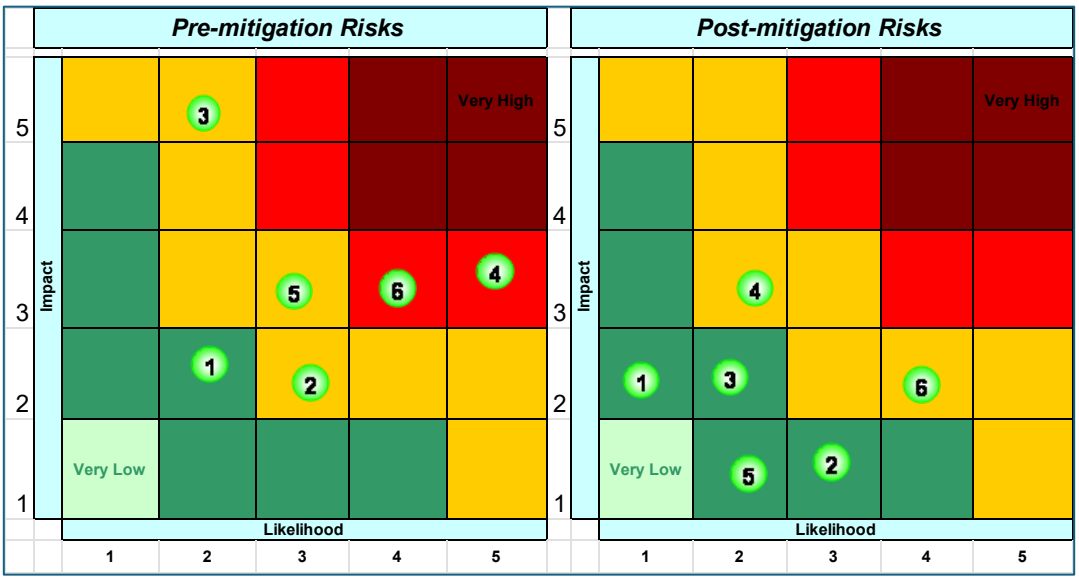

Risk ID		Risk Description	Pre-mitigation I	Pre-mitigation L	Pre-mitigation Score		Mitigation	Post-mitigation I	Post-mitigation L	Post-mitigation Score	
1	Risk (Event)	**Critical resource illness**	2	2	4	Low	Have defined alternate for tasks. Communicate potential issues as early as possible	2	1	2	Low
	Cause(s)	General risk of life with moderate-high external interactions									
	Effect(s)	Could cause slip in scheduled tasks, might require additional work from other resources									
2	Risk (Event)	**Laptop failure**	2	3	6	Medium	Computationally intensive tasks carried out on more robust laptops. Complete any remaining work on university computer	1	3	3	Low
	Cause(s)	Model complexity freezes inadequate hardware, old hardware, or physical damage									
	Effect(s)	Potential to cause misses for key milestones, likely to impact critical path									
3	Risk (Event)	**Data loss**	5	2	10	Medium	Frequent and consistent use of established GitHub. Backing up/saving data regularly.	2	2	4	Low
	Cause(s)	Numerous hardware/software errors possible.									
	Effect(s)	Loss of significant time investment on work. Potential to cause misses for key milestones									
4	Risk (Event)	**Model too complex**	3	5	15	High	Create a framework for a relatively simple minimum viable product. Scrap features that are the least necessary and have the highest computational burden	3	2	6	Medium
	Cause(s)	Overly ambitious design choices, inadequate grasp of the full data and application of the data									
	Effect(s)	Requires significant reworking of previous efforts. Delays progress on key milestones.									
5	Risk (Event)	**Not targetted enough answer**	3	3	9	Medium	Conduct and review stakeholder analysis. Well defined scope in project plan. Avoid scope creep through frequent reviews during workshops.	1	2	2	Low
	Cause(s)	Difficulty avoiding tunnel vision on personal interests. Challenges of problem formulation.									
	Effect(s)	Possible poor result for customer. Requires team-wide effort to refocus tasks.									
6	Risk (Event)	**Competing resource priorities**	3	4	12	High	Well defined alternatives for every task. No disproportionate spread of responsibility for tasks.	2	4	8	Medium
	Cause(s)	Every resource is taking 4-5 additional courses, assignments tend to have frequent overlap on delivery windows.									
	Effect(s)	Forces more work onto other resources. Potential impact to project schedule.									

Appendix:

[1] Lamas Fernandez, C., Matrinez-Sykora, A., Potts, C. (2021), Scheduling Double Round-Robin Sports Tournaments, ITC2021.

[2] Aygül, Ö., Hellgren, T., Azizi, S., & Trapp, A. C. (2025), A predict-and-prescribe framework for dynamic course scheduling toward strategic university scaling, *Omega*, 103406.

[3] Chen, R., Shih, H. (2013). Solving University Course Timetabling Problems Using Constriction Particle Swarm Optimization with Local Search, Algorithms, 6(2), Available at: Solving University Course Timetabling Problems Using Constriction Particle Swarm Optimization with Local Search (Accessed 02/02/2026).

[4] Alghamdi, H., Alsubait, T., Alhakami, H., Baz, A. (2021), A Review of Optimization Algorithms for University Timetable Scheduling, Engineering, Technology & Applied Science Research, 10(6), Available at: A Review of Optimization Algorithms for University Timetable Scheduling | Engineering, Technology & Applied Science Research (Accessed 03/02/2026).